

Modeling and Simulation of the Neonatal Respiratory Control System

Biological System Modeling Course Project

1 Introduction

Modeling and simulation of biological systems serve as indispensable tools for analyzing physiological phenomena and investigating the effects of various parameters on systemic performance. Given that this research focuses on the respiratory control system of newborn infants—and biological processes inherently exhibit nonlinear dynamics, time-dependent variables, and complex interrelations—simulation environments such as **MATLAB/Simulink** enable a rigorous understanding and practical assessment of models proposed in the scientific literature.

In this project, the model presented in the reference paper is implemented in the **Simulink** environment. To achieve this, the overall architecture was first examined, and all governing differential equations and parameters were subsequently mapped into appropriate functional blocks. The constituent modules were interconnected using fundamental and custom function blocks to faithfully reproduce the dynamical behavior described in the reference literature. Following model construction, simulation conditions and initial states were configured to align closely with the published benchmark data.

The primary objectives of this project are:

- Precise reconstruction of the mathematical model and block diagrams from the reference paper in Simulink.
- Evaluation and validation of system responses across diverse physiological conditions (resting state, hypoxia, and hypercapnia).
- Acquiring a deeper insight into the mathematical formulation and nonlinear dynamics of the neonatal respiratory system.
- Demonstrating the capabilities of the **MATLAB/Simulink** environment in analyzing complex bioelectrical and physiological systems.



Figure 1: Overall conceptual model of the neonatal respiratory system.

2 Concordance Between the Simulator Architecture and the Reference Model

This section examines the structural alignment between the block diagram implemented in **MATLAB/Simulink** and the conceptual model introduced in the reference paper. Figure ?? illustrates the original block diagram from the literature alongside its implemented Simulink counterpart. To preserve physiological integrity, the complete model is decoupled into two primary subsystems separated by a continuous horizontal boundary.

2.1 Controller Subsystem (Brainstem)

In the upper tier, the comprehensive `Discrete_Controller` block consolidates the functions of five discrete modules from the reference model. This unification enhances numerical stability and optimizes the execution of discrete-time state variables. These modules comprise:

- Mean value detector
- Clock pulse generator
- Frequency optimizer
- Ventilation controller
- Signal generator

2.2 Plant Subsystem (Body Organs)

In the lower tier, the differential equations governing gas exchange and systemic circulation are realized via four embedded **MATLAB** functions:

- **Lungs:** Receives the airflow rate demand (dv/dt) and supplies oxygenated arterial blood to systemic organs.

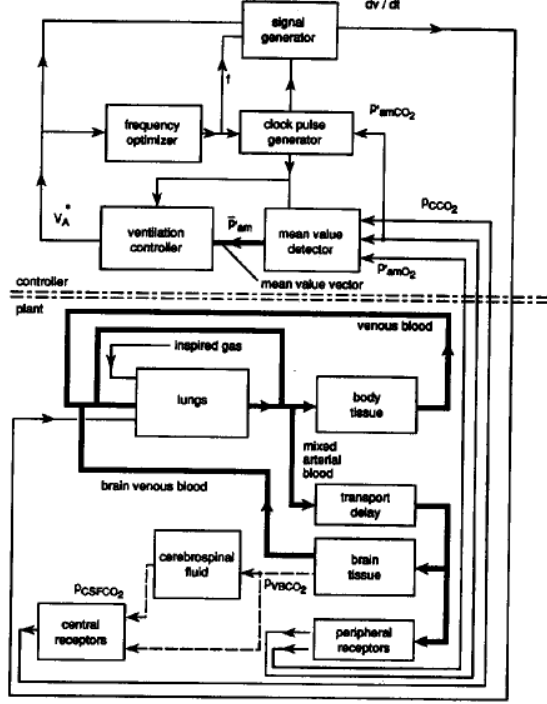


Figure 2: Conceptual block diagram from the paper.

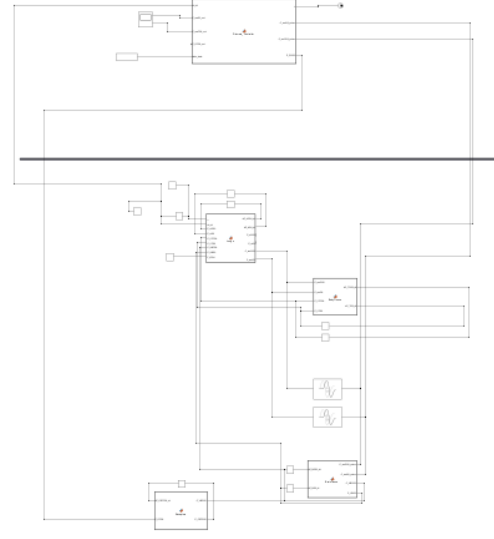


Figure 3: Model implemented in Simulink.

Figure 4: Structural comparison between the reference model and the simulator.

- **Body Tissue:** Consumes oxygen, produces metabolic carbon dioxide, and returns mixed venous blood directly to the lungs.
- **Brain Tissue:** Undergoes gas exchange with arterial blood and transfers cerebral venous outflow toward the cerebrospinal fluid chamber.
- **Cerebrospinal Fluid (CSF):** The bottom-most compartment, responsible for generating the central feedback chemical drive (P_{cCO_2}) that stimulates central chemoreceptors.

Note: To maximize computational efficiency, elements such as peripheral chemoreceptors and circulatory transport delays are integrated directly into the mathematical formulations of the core blocks.

3 Equation Implementation in Simulink

To implement the model in Simulink, the governing differential equations must be rewritten into explicit state-derivative form, isolating the first-order time derivatives on the left-hand side. These expressions are embedded within **MATLAB Function** blocks, and their calculated derivatives are routed to numerical integrator blocks for dynamic state integration.

As an illustrative example, the derivation for the **Body Tissue** compartment is detailed below. After isolating state derivatives, the mass-balance formulations assume the following standard form:

$$\frac{dC_{TCO_2}}{dt} = \frac{C_{amCO_2}Q_T + MR_{TCO_2} - C_{VTCO_2}Q_T}{S_T} \quad (1)$$

$$\frac{dC_{TO_2}}{dt} = \frac{C_{amO_2}Q_T - MR_{TO_2} - C_{VTO_2}Q_T}{S_T} \quad (2)$$

Where the parameters are physiologically defined as:

- Q_T : Cardiac output (blood flow rate)
- MR : Metabolic rate of tissue gas consumption or production
- S_T : Effective tissue storage volume factor
- C_{am} and C_{VT} : Mixed arterial and venous gas concentrations, respectively

The programmatic code corresponding to these differential equations was implemented in MATLAB as follows, with physiological parameters taken directly from the paper's appendix:

MATLAB Function – Body Tissue

```
function [dC_TCO2_dt, dC_TO2_dt] = BodyTissue(C_amCO2, C_amO2,
C_VTCO2, C_VTO2)
% --- Constants from Appendix ---
Q_T = 9.8e-3;           % (l/s)
MR_TCO2 = 1.625e-4;    % (l/s)
MR_TO2 = 1.902e-4;    % (l/s)
S_T = 0.9;             % (l)

% --- Equations ---
dC_TCO2_dt = (C_amCO2 * Q_T + MR_TCO2 - C_VTCO2 * Q_T) / S_T;
dC_TO2_dt = (C_amO2 * Q_T - MR_TO2 - C_VTO2 * Q_T) / S_T;
end
```

As illustrated, the outputs of the MATLAB Function blocks feed into integrator blocks for numerical solving, and the resolved states loop back into the compartment or drive interconnected physiological subsystems.

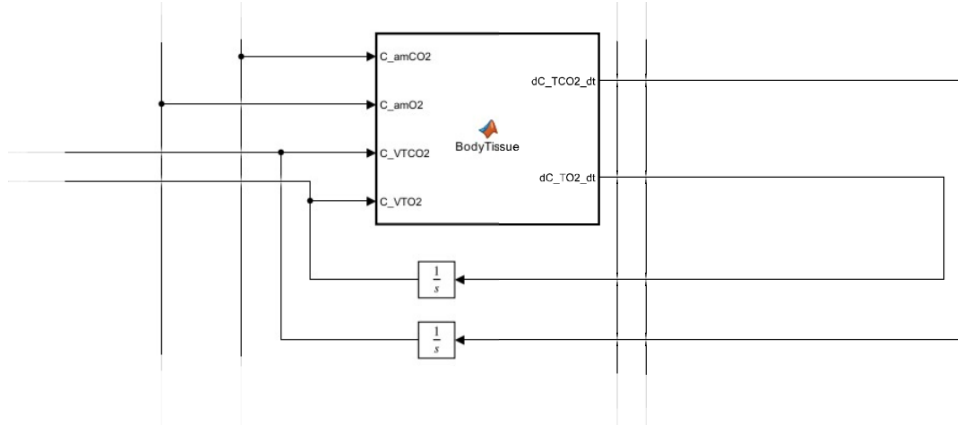


Figure 5: Interconnection and signal routing of the Body Tissue subsystem.

4 Simulation Results and Discussion

In this section, model outputs across three distinct physiological regimes are evaluated and compared against findings reported in the reference literature. To ensure complete stabilization, all state trajectories were recorded after 8 minutes of simulation time, representing the true steady-state.

Test Condition			Simulation Results						
F_{IO_2}	F_{ICO_2}	μ	P_{amO_2}	P_{amCO_2}	P_{cCO_2}	V_A	V_D	V_E	f1
Resting Conditions									
0.21	0	1.0	73	39.4	41.8	0.491	3.70×10^{-3}	0.6	30.0
Hypoxia									
0.15	0	1.0	48.4	35.5	39.1	0.623	4.04×10^{-3}	0.754	32.5
0.15	0	0.8	47.9	36.5	39.4	0.614	4.02×10^{-3}	0.743	32.3
0.15	0	0.5	46.1	36.8	40.1	0.481	3.64×10^{-3}	0.589	29.8
0.12	0	1.0	38.1	30.3	35.4	0.806	4.56×10^{-3}	0.965	35.0
0.12	0	0.8	36.5	31.3	36.2	0.778	4.48×10^{-3}	0.933	34.7
0.12	0	0.5	34.2	33	37.6	0.649	4.12×10^{-3}	0.784	32.88
Hypercapnia									
0.21	0.03	1.0	98.8	40.8	43.4	0.848	4.68×10^{-3}	1.014	35.5
0.21	0.03	0.8	98.7	40.8	43.4	0.848	4.68×10^{-3}	1.014	35.5
0.21	0.03	0.5	98.6	40.8	43.4	0.849	4.68×10^{-3}	1.015	35.5
0.21	0.05	1.0	106.6	44.8	46.2	1.869	7.57×10^{-3}	2.187	42.1
0.21	0.05	0.8	106.7	44.5	46.1	1.826	7.45×10^{-3}	2.138	41.9
0.21	0.05	0.5	106.7	44.3	46.2	1.804	7.38×10^{-3}	2.114	41.8

Figure 6: Steady-state benchmark values reported in the reference paper (Table 1).

The parametric trends (such as the sharp rise in minute ventilation upon hypoxic exposure) exhibit strong concordance with the reference study. Minor discrepancies in certain absolute values (particularly oxygen tensions) arise from differences in solver precision between modern stiff variable-step algorithms (`ode45` in Simulink) and historical codebases, alongside the high numerical sensitivity of the oxygen-hemoglobin dissociation curve at elevated pressures.

4.1 Resting Conditions

The model was initially evaluated under baseline resting conditions ($F_{IO_2} = 21\%$ and $F_{ICO_2} = 0\%$). Figure ?? displays the total respiratory airflow signal, where a peak

Table 1: Steady-state simulation results (after 8 minutes) across resting, hypoxia, and hypercapnia conditions (current model output).

Test Condition			Simulation Results						
$F_{I_{O_2}}$	$F_{I_{CO_2}}$	μ	P_{amO_2}	P_{amCO_2}	P_{cCO_2}	V_A	V_D	V_E	f
<i>Resting Conditions</i>									
0.21	0	1.0	78.0	38.8	43.7	0.478	3.63×10^{-3}	0.589	30.5
<i>Hypoxia</i>									
0.15	0	1.0	51.7	36.0	41.8	0.560	3.86×10^{-3}	0.684	32.2
0.15	0	0.8	50.9	36.5	42.1	0.563	3.87×10^{-3}	0.688	32.3
0.15	0	0.5	49.7	37.2	42.6	0.542	3.81×10^{-3}	0.663	31.9
0.12	0	1.0	40.6	31.9	38.8	0.780	4.49×10^{-3}	0.940	35.7
0.12	0	0.8	39.5	32.7	39.5	0.732	4.35×10^{-3}	0.884	35.0
0.12	0	0.5	37.4	34.4	40.7	0.666	4.17×10^{-3}	0.808	34.1
<i>Hypercapnia</i>									
0.21	0.03	1.0	88.5	41.3	45.5	0.913	4.87×10^{-3}	1.095	37.3
0.21	0.03	0.8	88.6	41.3	45.5	0.924	4.90×10^{-3}	1.107	37.4
0.21	0.03	0.5	88.6	41.3	45.6	0.921	4.89×10^{-3}	1.104	37.3
0.21	0.05	1.0	91.9	45.0	48.5	1.604	6.82×10^{-3}	1.891	42.2
0.21	0.05	0.8	92.2	45.2	48.5	1.632	6.90×10^{-3}	1.924	42.3
0.21	0.05	0.5	92.5	45.3	48.6	1.666	6.99×10^{-3}	1.963	42.5

oscillatory amplitude near 0.04 l/s demonstrates high fidelity to the reference waveforms.

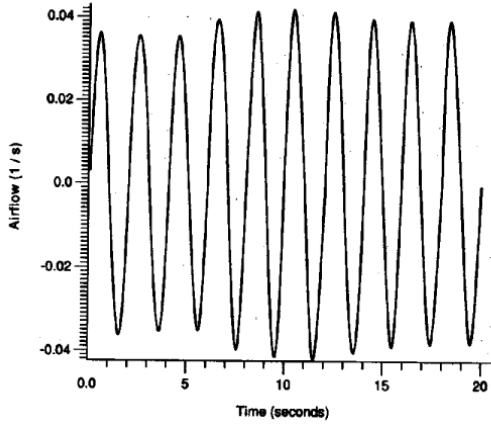


Figure 7: Reference respiratory airflow.

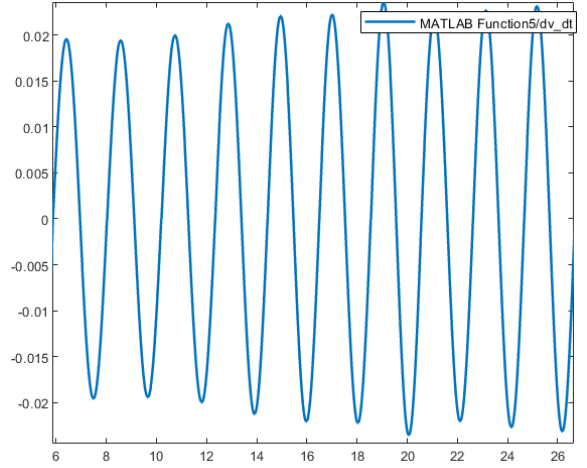


Figure 8: Simulated model airflow (present study).

Figure 9: Comparison of respiratory airflow under resting conditions.

In addition to airflow dynamics, the stabilization trajectories of arterial blood gas tensions were tracked (Figure ??). By optimizing tidal volume and frequency, the controller drives P_{amO_2} and P_{amCO_2} to stable levels within physiological norms following an initial transient phase.

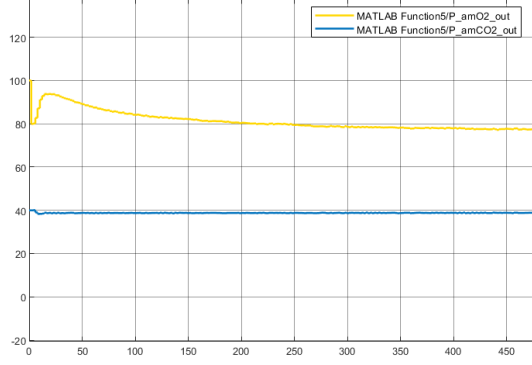


Figure 10: Stabilization trajectories of blood gas partial pressures toward resting equilibrium.

4.2 Hypoxia Conditions

The response of the system to environmental oxygen deprivation was tested by stepping down inspired oxygen fractions to 15% and 12%.

15% Hypoxia: Following the step drop in ambient oxygen (Figure ??), the brainstem compensates for reduced arterial oxygen (P_{amO_2}) by driving hyperventilation. This leads to increased CO_2 clearance from the lungs and a slight decline in arterial carbon dioxide (P_{amCO_2}), accurately reproducing the documented physiological mechanism.

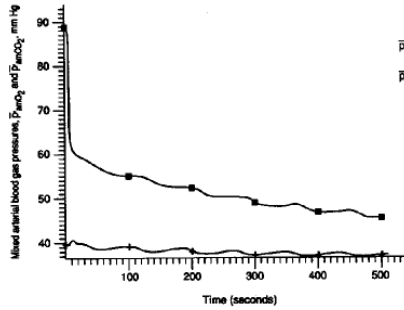


Figure 11: Reference trajectory for 15% hypoxia.

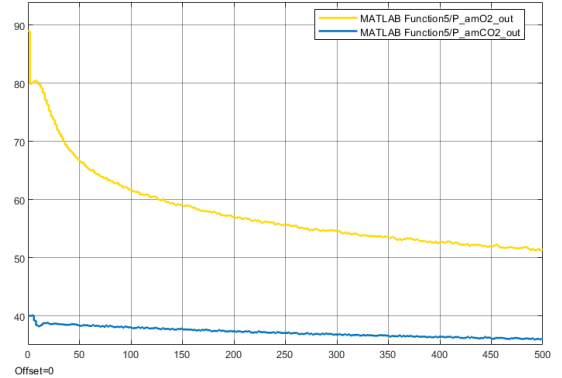
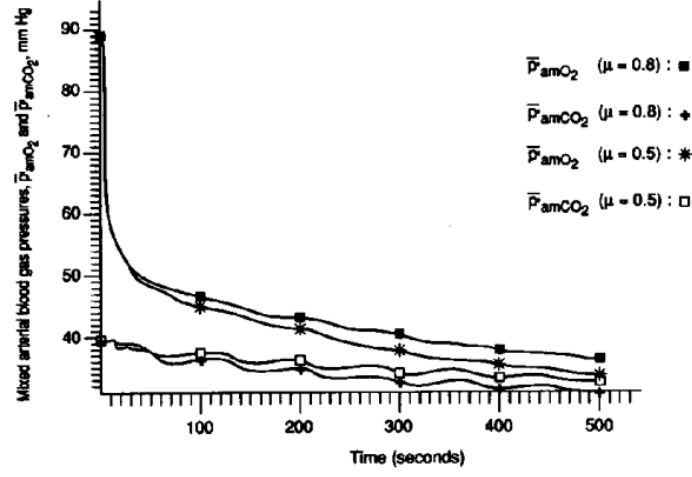


Figure 12: Model output for 15% hypoxia ($\mu = 0.5$).

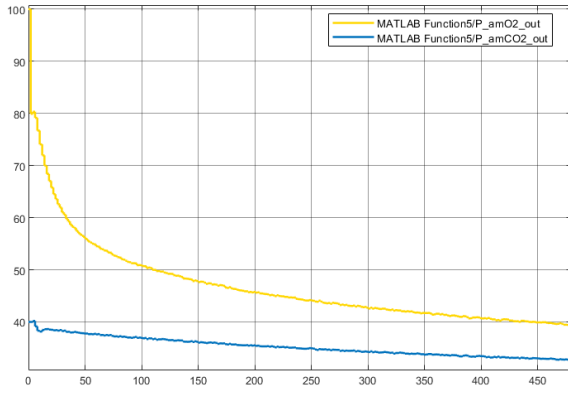
Figure 13: Dynamic response comparison under 15% hypoxia.

12% Hypoxia and the Role of Weighting Parameter μ : To determine the sensitivity of the frequency optimizer to mechanical work weighting (μ), the trial was repeated at 12% O_2 using $\mu = 0.8$ and $\mu = 0.5$ (Figure ??). Lowering μ to 0.5 reduces the penalty on chemical error relative to breathing work, stabilizing oxygen at a slightly lower level. This response confirms the correct numerical operation of the frequency optimization algorithm.

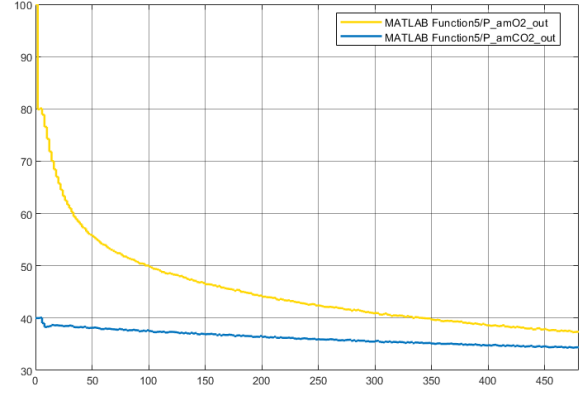
Figure ?? illustrates the corresponding rise in respiratory flow amplitude, demonstrating the neural mechanism that increases tidal volume to combat oxygen depletion.



(a) Reference 12% hypoxia trajectory.



(a) Model output ($\mu = 0.8$).



(b) Model output ($\mu = 0.5$).

Figure 15: Evaluation of weighting parameter μ under 12% hypoxia.

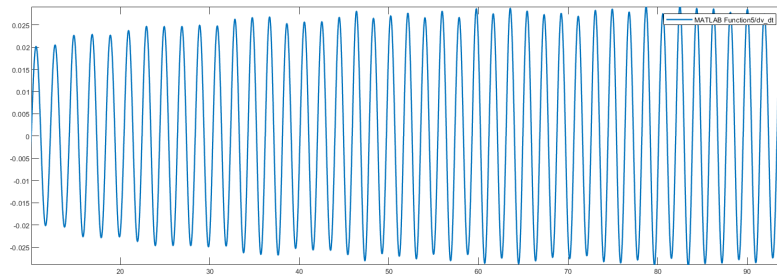


Figure 16: Compensatory increase in respiratory depth during acute hypoxia.

4.3 Hypercapnia and Time-Varying Dynamics

To study transient system behavior under carbon dioxide loading, the model was initially brought to resting steady-state. From $t = 0$ s to $t = 500$ s, an inspired mixture containing 5% CO_2 was delivered, followed by an atmospheric wash-out recovery phase (Figure ??).

Stimulation Phase (0 to 500 seconds): The entry of CO_2 into the lungs triggers vigorous neural hyperventilation. In the presence of 21% ambient oxygen, this extreme ventilation drives P_{amO_2} above 90 mmHg. Simultaneously, arterial CO_2 rises with a moderate slope to roughly 45 mmHg despite ongoing compensation.

Recovery Phase (after 500 seconds): Following the removal of CO_2 , elevated ventilation rapidly exhausts excess accumulated blood gas, resulting in a precipitous drop in P_{amCO_2} . After this steep decline, the controller undergoes brief damped oscillations before resetting to resting baseline.

Comparative assessment reveals that the dual-phase transient dynamics during both stimulation and recovery are accurately reproduced.

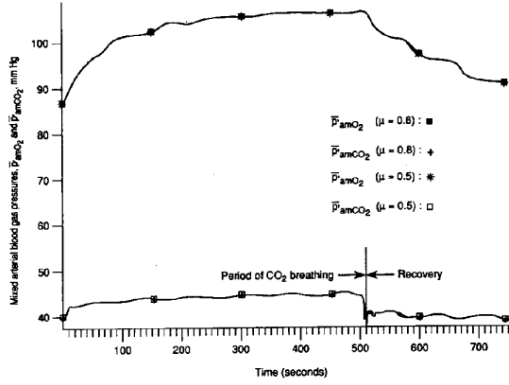


Figure 17: Response to 5% CO_2 (Reference).

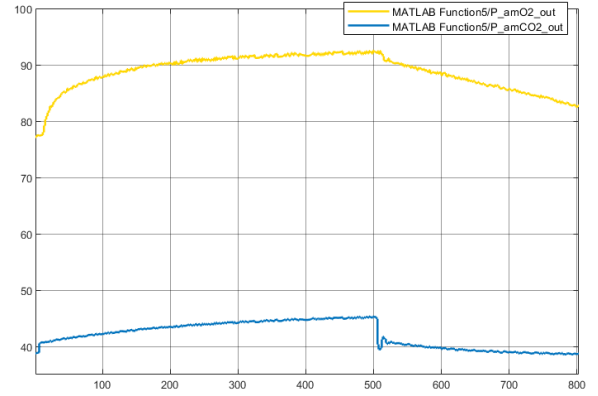


Figure 18: Response to 5% CO_2 (Current Model).

Figure 19: Comparison of time-varying hypercapnia and wash-out recovery dynamics ($\mu = 0.8$).

5 Conclusion

In this study, the mathematical model of the neonatal respiratory control system was successfully re-implemented and validated in the **MATLAB/Simulink** environment. By translating dynamic differential equations into state-space formulations coupled with modern numerical ODE solvers, the intricate biological dynamics were encapsulated within an integrated computational framework.

System evaluation across diverse physiological states demonstrated strong robustness against environmental disturbances. Under hypoxic stress, the controller compensated for blood oxygen desaturation by elevating breathing depth, verifying the critical role of the work-minimization parameter (μ). In hypercapnic conditions, the model captured dual-phase, time-varying behaviors across both active challenge and recovery regimes. Overall, the close agreement between simulation outputs and literature benchmarks confirms that block-based Simulink implementations provide an effective platform for analyzing, troubleshooting, and extending complex physiological systems.